Classical Problems Characterizing the Power of Quantum Computing

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Talk is based on

- A single-shot measurement of the energy of product states in a translation invariant spin chain can replace any quantum computation, `quant-ph/0710.1615`
- A Simple PromiseBQP-complete Problem, *Theory of Computing*, Volume 3, Article 4, 2007
- A PromiseBQP-complete String Rewriting Problem, `quant-ph/0705.1180`

Challenges in Quantum Computer Science

- designing new efficient quantum algorithms
- understanding the computational power and limitations of quantum computers in comparision to those of classical computers

Complexity Class PromiseBQP

PromiseBQP is the class of promise problems $(\Pi_{\mathrm{YES}}, \Pi_{\mathrm{NO}})$ that can be solved by a *uniform* family of quantum circuits $\{U_n\}$ in the following sense:

Let $\mathbf{x}$ be a binary string of length n. Then the application of U_n to $|\mathbf{x}, \mathbf{0}\rangle$ produces the state

$$U_n|\mathbf{x}, \mathbf{0}\rangle = \alpha_{\mathbf{x},0}|0\rangle \otimes |\psi_{\mathbf{x},0}\rangle + \alpha_{\mathbf{x},1}|1\rangle \otimes |\psi_{\mathbf{x},1}\rangle$$

such that

- $|\alpha_{\mathbf{x},1}|^2 \geq 2/3$ if $x \in \Pi_{\mathrm{YES}}$
- $|\alpha_{\mathbf{x},1}|^2 \leq 1/3$ if $x \in \Pi_{\mathrm{NO}}$

Measurements of Observables

- a measurement of the obervable A with spectral decomposition

$$A = \sum_{\lambda} \lambda \, Q_{\lambda}$$

is a quantum process that

- outputs the eigenvalue λ
- with probability $\mathrm{tr}(Q_{\lambda} \, \rho)$

when applied to an arbitrary quantum state ρ

Remarks: Measurements of Observables

- note that
  - the post-measurement state is irrelevant
  - a quantum process with the property that the expectation of the generated values is equal to $\mathrm{tr}(A\,\rho)$ may not suffice

Spectral measure

- the spectral measure induced by the observable A and the state ρ is the probability distribution on $\Omega = \mathbb{R}$ where

$$\Pr(\lambda) = \mathrm{tr}(Q_\lambda \rho)$$

  we set $Q_\lambda = \mathbf{0}$ for all $\lambda \notin \mathrm{Spec}(A)$
- a measurement of A in the state ρ allows us to sample according the spectral measure

Accuracy of Measurements

We say that an approximate measurement of A has

- maximal error δ and
- reliability $1 - \epsilon$

iff

$$\Pr(\lambda \in [a - \delta, b + \delta]) \geq \mathrm{tr}(Q_{[a,b]}\, \rho) \cdot (1 - \epsilon)$$

for all intervals $[a, b] \subseteq \mathbb{R}$ and all states ρ

where $Q_{[a,b]} = \sum_{\mu \in [a,b]} Q_\mu$

Efficient Implementation of Measurements

- let A be a sparse matrix of size N
- a measurement of A with maximal error δ and reliability $1-\epsilon$ can be implemented with resources (time and space) that are polynomial in $1/\delta$, $1/\epsilon$ and $\log N$
- the idea is to
  - simulate time evolution $U \approx \exp(-iA/\|A\|)$
  - apply phase estimation to U

Quantum Computution with Measurements

- there is a family of Hamiltonians $H^{(m)}$ acting on qudit chains $(\mathbb{C}^{60})^{\otimes m}$

$$H^{(m)} = \sum_{j=0}^{m-2} H_{j,j+1}$$

where $m = poly(n)$ such that a single-shot measurement of the energy of

- canonical basis states
- with error $\delta = 1/poly(m)$ and reliability $1 - \epsilon$

solves BQP in a probabilistic sense

Properties of the Hamiltonians

- the Hamiltonians $H^{(m)}$ are “physically realistic” since
  - each $H^{(m)}$ is translation invariant
  - each $H^{(m)}$ consists of nearest-neighbor interactions; H acts on $(\mathbb{C}^{60})^{\otimes 2}$

Solving BQP in a probabilistic sense

- for all input strings $\mathbf{x}$ of length n there is
  - a partition $\mathbb{R} = \mathcal{Y}_m \dot{\cup} \mathcal{N}_m$ and
  - a canonical basis state $|\psi_x\rangle \in (\mathbb{C}^{60})^{\otimes m}$ such that

$$\Pr(\lambda \in \mathcal{Y}_m \,|\, x \in \Pi_{\mathrm{YES}}) \quad \geq \quad p_{\mathbf{x},1}(1-\epsilon)$$

$$\Pr(\lambda \in \mathcal{N}_m \,|\, x \in \Pi_{\mathrm{NO}}) \quad \geq \quad p_{\mathbf{x},0}(1-\epsilon)$$

  where λ is the random output of the approximate energy measurement of $H^{(m)}$ in the state $|\psi_x\rangle$

- the objects m, $|\psi_x\rangle$, Y_m, and N_m can be efficiently computed from x and $U^{(n)}$

Idea of the Proof

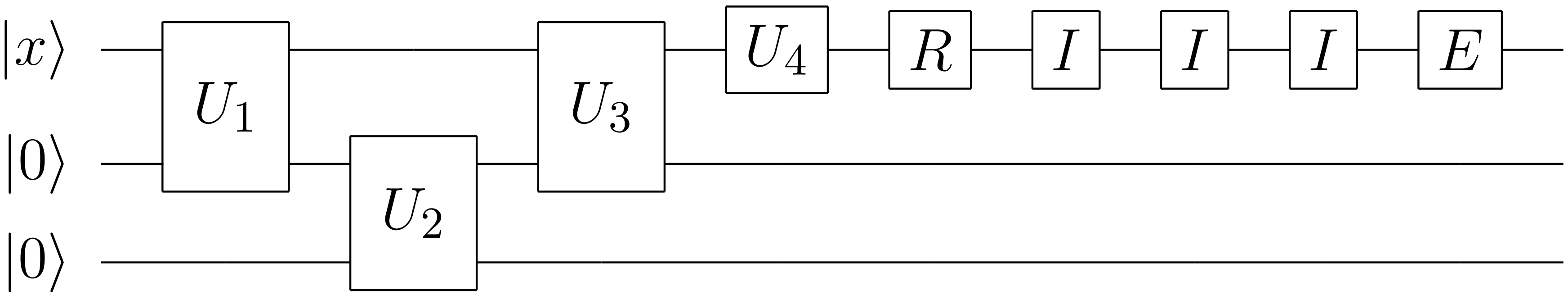
- let $U = U_r U_{r-1} \cdots U_1$ be an arbitrary quantum circuit
- define the non-unitary gates $R := |1\rangle\langle 1|$ and $E := \mathbf{0}$
- for $s > r$ define the non-unitary circuit

$$\hat{U} = \hat{U}_{s+1}\hat{U}_s\hat{U}_{s-1}\cdots\hat{U}_{r+1}\hat{U}_r\hat{U}_{r-1}\cdots\hat{U}_1$$

  - the gates of $\hat{U}$ are

$$\begin{aligned}
\hat{U}_t &:= U_t && \text{for } t = 1,\ldots,r \\
\hat{U}_{r+1} &:= R && \\
\hat{U}_t &:= I && \text{for } t = r+2,\ldots,s \\
\hat{U}_{s+1} &:= E &&
\end{aligned}$$

Example



- $r = 4$ and $s = 8$
- read-out gate $R := |1\rangle\langle 1|$ and end-gate $E := \mathbf{0}$

Forward-Time Operator

- adjoin a Hilbert space $\mathcal{H}_{\text{clock}}$ to the quantum register
- define the forward operator F by

$$\begin{aligned} F\Big(|\varphi\rangle \otimes |t\rangle\Big) &= \hat{U}_{t+1}|\varphi\rangle \otimes |t+1\rangle \text{ for } t = 0, \ldots, s \\ F^{\dagger}\Big(|\varphi\rangle \otimes |0\rangle\Big) &= 0 \end{aligned}$$

- define the initial state $|\psi\rangle := |\mathbf{x}, \mathbf{0}\rangle \otimes |0\rangle_{\text{clock}}$

Intuitive Action of F

- F operates as follows:
  - F implements $U = U_r U_{r-1} \cdots U_1$
  - F checks the output of U (with read-out gate R)
  - output NO $\Rightarrow F$ stops
  - output YES $\Rightarrow F$ carries out idle steps
  - F stops unconditionally (end-gate E)

Action of F

- define the matrix $\hat{F}$ acting on $\mathcal{H} := \mathbb{C}^{r+1} \oplus \mathbb{C}^{s+1}$ by

$$\hat{F} := S_{r+1} \oplus S_{s+1}$$

  S_{r+1} and S_{s+1} are the adjacency matrices of line graphs with $r+1$ and $s+1$ vertices

- define the state $|\phi\rangle := \sqrt{1-p_1}|0\rangle \oplus \sqrt{p_1}|0\rangle$

- $\Rightarrow$ there is a nice isomorphism between F and $\hat{F}$

$$\mathrm{span}\{F^t|\psi\rangle\} \;\cong\; \mathcal{H}$$

$$F^t|\psi\rangle \;\cong\; \hat{F}^t|\phi\rangle$$

Spectral measure

- $\Rightarrow$ the spectral measure induced by

$$H = F + F^\dagger \text{ and } |\psi\rangle = |\mathbf{x}, \mathbf{0}\rangle \otimes |0\rangle_{\text{clock}}$$

  is a convex combination

$$(1 - p_1)P_{r+1} + p_1 P_{s+1}$$

  where P_{r+1} and P_{s+1} are probability distributions on the spectra of the line graphs with $r+1$ and $s+1$ vertices

- the spectra $\mathcal{S}_{r+1}$ and $\mathcal{S}_{s+1}$ are disjoint and the distance between them is $\Omega(1/poly(r))$

- $\Rightarrow$ quantum computation can be replaced by a single-shot approximate energy measurement

A Simple PromiseBQP-complete Matrix Problem

- let A be a sparse matrix of size N
- given $m = polylog(N)$, $j \in [1, \ldots, N]$, $\epsilon = 1/polylog(N)$, and $g \in [-\|A\|^m, \|A\|^m]$
  decide if
  - $(A^m) >$

A PromiseBQP-complete String Rewriting Problem

- let $\mathcal{A}$ be a finite alphabet
- let $\sim$ be a symmetric relation on substrings of length at most k (where k is a constant)
- let A be the adjacency matrix of the undirected graph $G = (V, E)$, where
  - the set of vertices V is $\mathcal{A}^r$ (strings of length r over $\mathcal{A}$)
  - the set of edges E corresponds to possible conversions between strings, that is,

$$(avb, awb) \in E \quad \Longleftrightarrow \quad v \sim w$$

String Rewriting Problem

- let s, t, and t' be three strings in $\mathcal{A}^r$
- let
$$\Delta_{s,t,t'}(n) := A^n_{s,t} - A^n_{s,t'}$$
be the difference between the number of possibilities to obtain t from s and those to obtain t' from s after exactly n replacements
- given the promise
  - $|\Delta_{s,t,t'}(n)| \leq c^n$ for all $n \in \mathbb{N}$ (*growth condition*)
  - $|\Delta_{s,t,t'}(m)| \geq \epsilon c^m$

  determine the sign of $\Delta_{s,t,t'}(m)$ where
  $c > 0$ is a constant, $m \in \mathbb{N}$ with $m = poly(r)$, and $\epsilon \in [0, 1]$ with $\epsilon = poly(1/r)$

Motivation from Bioinformatics

- strings s, t, and t' correspond to DNA sequences
- mutations change substrings (possible mutations are specified by the relation $\sim$)
- String rewriting problem:
  Are there more possibilities to mutate from s to t or from s to t' if exactly m mutations occur?

PromiseBQP-complete

Theorem: String Rewriting is PromiseBQP-complete

The string rewriting problem is the most difficult problem that can be efficiently solved on a quantum computer.

Quantum computers are better at estimating differences of combinatorial quantities than classical computers (provided that BPP is not equal to BQP).

Conclusions

“Spectral” Quantum Algorithms

- Approximate evaluation of Jones, HOMFLYPT, and Tutte polynomials (PromiseBQP-complete problems)
- Evaluation of NAND-Trees